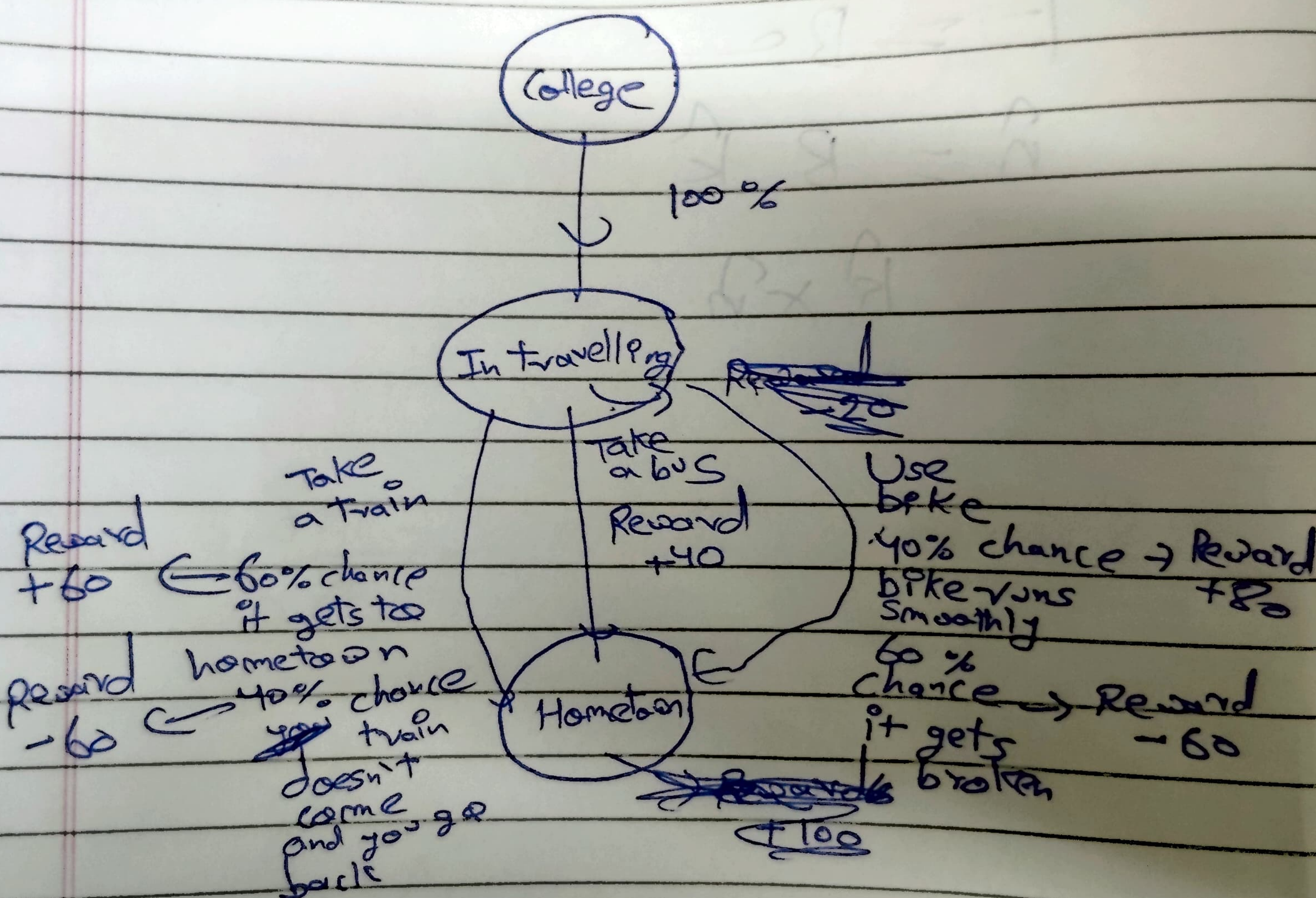
MDP

This MDP is about travelling from college to my hometown



MDP

This is an MDP based on going from college to hometown,

State space -: S_1 : college
S_2 : In travelling
S_3 : Hometown

Action space -: (In state S_2)

Transition Probabilities

A_1 : Take a train

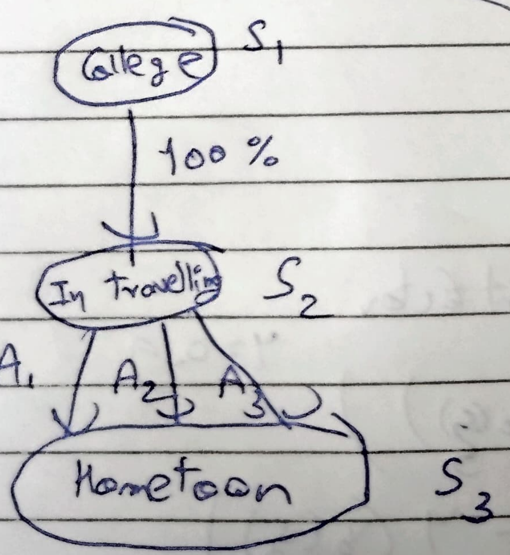
Reward → +60 → 0.6 chance it gets to hometown

Reward → -60 → 0.4 chance it stays in S_2

A_2 : Take a bus → 100% chance goes to S_3
Reward = +40

A_3 : Take a bike → 0.4 chance goes to S_3
Reward = +80

→ 0.6 chance stays in S_2
Reward = −80



~~[illegible]~~

Initializing values

$V_0(s_1)=0$ $V_0(s_2)=0$ $V_0(s_3)=0$

First step

For state s_1

~~Using bellman Eqn~~

~~$V_1(s_1)= \sum P($~~

value of remaining at college

$V_1(s_1)=0$

As 100 % we are going to S_2

For state s_3

value of remaining at hometown as it is the absorbing state

$V_1(s_3)=0$

For state s_2

~~$V_1(s_2)=0.6$~~ $[$

For A_1, discount factor

$\gamma=0.9$

$$\text{Expected value}=0.6\left(60+\gamma\left(V_0(s_3)\right)\right)+0.4\left(-60+\gamma\left(V_0(s_2)\right)\right)$$

$$=0.6\left(60+0.9(0)\right)+0.4\left(-60+0.9(0)\right)$$

$$=12$$

For A_2

$$\text{Expected value} = 1 \times [40 + 0.9(0)] = 40$$

For a_3

$$\text{Expected value} = 0.4[80 + 0.9(0)] + 0.6[-60 + 0.9(0)]$$
$$= -4$$

Using bellman equation $V_1(S_2) = Max(12, 40, -4) = 40$

Second step

For state S_3

$$V_2(S_3) = 0$$

For state S_1

$$V_2(S_1) = 0$$

For state S_2

For action A_1

$$\text{Expected value} = 0.6(60 + 0.9(0)) + 0.4(-60 + 0.9(40))$$
$$= 36 + 0.4(-60 + 36)$$
$$= 36 + 0.4(-24)$$
$$= 36 - 9.6 = 26.4$$

For action A_2

$$= 1 \times [40 + 0.9(0)] = 40$$

For action A_3

$$0.4[80 + 0.9(0)] + 0.6[-60 + 0.9(40)]$$
$$= 32 + 0.6(-60 + 36) = 32 + 0.6(-24) = 32 - 14.4 = 17.6$$

$\boxed{V_2(S_2) = 40}$